

# MI LDC API

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**Version 3.2**

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## REVISION HISTORY

| Revision No. | Description                  | Date       |
|--------------|------------------------------|------------|
| 3.0          | Initial release              | 12/04/2020 |
| 3.1          | Add MI_LDC_SetOutputPortAttr | 07/04/2021 |
|              | Add procfs introduction      | 08/25/2021 |
| 3.2          | Add Maruko info              | 04/01/2022 |
|              | Add new interface:           |            |
|              | 1. MI_LDC_AttachToChn        |            |
|              | 2. MI_LDC_DetachFromChn      |            |

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## 1. OVERVIEW

### 1.1. Module Description

LDC (Lens Distortion Correction), support Lens Distortion Correction、Image Stitch 、Blending、Digital Image Stabilization (DIS) and Lookup Table Feature (LUT).

### 1.2. Flow Chart



Note:

1. 32X2 and 16X2 are the input image and the output image width and height alignment restrictions.

### 1.3. Platform Features

Functions supported in different platforms:

Device 0

| Platform | Feature | LDC            | Stitch and Blending | DIS | LUT |
|----------|---------|----------------|---------------------|-----|-----|
| Tiramisu |         | Y              | Y                   | Y   | N   |
| Muffin   |         | Y              | Y                   | Y   | Y   |
| Mochi    |         | Y              | Y                   | Y   | Y   |
| Maruko   |         | Y (Horizontal) | N                   | N   | N   |

Device 1

| Platform | Feature | LDC | Stitch and Blending | DIS | LUT |
|----------|---------|-----|---------------------|-----|-----|
| Muffin   |         | Y   | Y                   | Y   | Y   |

Features:

Lens distortion correction

According to lens calibration parameters, input image is subjected to distortion correction processing, and then output according to the projection mode and the region of interest configured by the user.

Image stitching and fusion

Distortion correction is performed on images come from multiple sensors, then multiple images are stitched

together, and finally the overlapping areas of the images are fused.

Digital image stabilization:

Currently supports anti-shake algorithm with gyroscope, that is, calculate the movement offset of the image of the current frame according to the data generated by the gyroscope, and then according to the motion offset condition performs translation, rotation and other transformations on the current image, so as to achieve the effect of anti-shake.

Lookup table

Provides a two-dimensional lookup table function. The data bit width of the index and table in the x and y directions are both 8bit, which can be used for fast search of the weight table of the image fusion algorithm.

## 1.4. Keyword

- Device  
Ldc hardware device
- Channel  
The LDC module processes channels, and the LDC hardware of each channel is time-sharing multiplexed, Currently only supports single channel.

## 2. API LIST

---

This module provides the following APIs:

| Name of API                              | Function                                                                         |
|------------------------------------------|----------------------------------------------------------------------------------|
| <a href="#">MI_LDC_CreateDevice</a>      | Create an LDC device                                                             |
| <a href="#">MI_LDC_DestroyDevice</a>     | Destroy an LDC device                                                            |
| <a href="#">MI_LDC_CreateChannel</a>     | Create an LDC channel                                                            |
| <a href="#">MI_LDC_DestroyChannel</a>    | Destroy an LDC channel                                                           |
| <a href="#">MI_LDC_StartChannel</a>      | Start an LDC channel                                                             |
| <a href="#">MI_LDC_StopChannel</a>       | Stop an LDC channel                                                              |
| <a href="#">MI_LDC_GetOutputPortAttr</a> | Get the LDC output attribute                                                     |
| <a href="#">MI_LDC_SetChnParam</a>       | Set the LDC channel attribute                                                    |
| <a href="#">MI_LDC_DoLutDirectTask</a>   | Execute table lookup tasks , send buffer directly to ldc hardware for processing |
| <a href="#">MI_LDC_SetOutputPortAttr</a> | Configure LDC output attribute                                                   |
| <a href="#">MI_LDC_AttachToChn</a>       | Bind Ldc Channel to another module channel                                       |
| <a href="#">MI_LDC_DetachFromChn</a>     | Unbind Ldc Channel                                                               |

### 2.1. MI\_LDC\_CreateDevice

➤ Description

Create an LDC device

➤ Syntax

MI\_S32 MI\_LDC\_CreateDevice(MI\_LDC\_DEV devId, [MI\\_LDC\\_DevAttr\\_t](#) \*pstDevAttr);

➤ Parameters

| Parameter Name | Description                   | Input/Output |
|----------------|-------------------------------|--------------|
| devId          | Device ID number              | Input        |
| pstDevAttr     | Device working mode attribute | Input        |

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

- Dependence
  - Header file: mi\_ldc.h
  - Lib: libmi\_ldc.a
- Note
  - None.
- Example  
LDC initialization settings and exit example:

Tiramisu, Muffin, Mochi Platform

```
MI_S32 s32Ret = MI_SUCCESS;
MI_LDC_DEV LdcDevid = 0;
MI_LDC_CHN LdcChnId = 0;
MI_LDC_DevAttr_t stDevAttr = {0};
MI_LDC_ChnAttr_t stChnAttr = {};
MI_LDC_SensorCalibInfo_t * pstCalibInfo = &stChnAttr.stCalibInfo;

stChnAttr.eMode = MI_LDC_WORKMODE_LDC ;
s32Ret = ReadBufFromFile("ldc.json", stChnAttr.pConfigAddr, stChnAttr.u32ConfigSize);
if (s32Ret)
{
    return s32Ret;
}

stChnAttr.eInfoType = MI_LDC_MAPINFOTYPE_SENSORCALIB;
s32Ret = ReadBufFromFile("CalibPoly.bin", pstCalibInfo.pCalibPolyBinAddr,
pstCalibInfo.u32CalibPolyBinSize);
if (s32Ret)
{
    return s32Ret;
}

MI_LDC_CreateDevice(LdcDevid, &stDevAttr);
MI_LDC_CreateChannel(LdcDevid, LdcChnId, &stChnAttr);
MI_LDC_StartChannel(LdcDevid, LdcChnId);

ReleaseBuf(stChnAttr.pConfigAddr);
ReleaseBuf(pstCalibInfo.pCalibPolyBinAddr);

/*Destroy channel*/
MI_LDC_StopChannel(LdcDevid, LdcChnId);
MI_LDC_DestroyChannel(LdcDevid, LdcChnId);
MI_LDC_DestroyDevice(LdcDevid);
```

Maruko Platform

```
MI_S32 s32Ret = MI_SUCCESS;
MI_LDC_DEV LdcDevid = 0;
MI_LDC_CHN LdcChnId = 0;
MI_LDC_DevAttr_t stDevAttr = {0};
MI_LDC_ChnAttr_t stChnAttr = {};
```



```
MI_SYS_ChnPort_t stDstChnPort = {};  
  
stChnAttr.eMode = MI_LDC_WORKMODE_LDC ;  
s32Ret = ReadBufFromFile("ldc.json", stChnAttr.pConfigAddr, stChnAttr.u32ConfigSize);  
if (s32Ret)  
{  
    return s32Ret;  
}  
  
stDstChnPort.eModId = E_MI_MODULE_ID_ISP;  
stDstChnPort.u32DevId = 0;  
stDstChnPort.u32ChnId = 0;  
stDstChnPort.u32PortId = 0;  
  
MI\_LDC\_CreateDevice(LdcDevid, &stDevAttr);  
MI\_LDC\_CreateChannel(LdcDevid, LdcChnId, &stChnAttr);  
MI\_LDC\_AttachToChn(LdcDevid, LdcChnId, &stDstChnPort);  
  
/*Exit and release buffer*/  
ReleaseBuf(stChnAttr.pConfigAddr);  
  
MI\_LDC\_DetachFormChn(LdcDevid, LdcChnId, &stDstChnPort);  
MI\_LDC\_DestroyChannel(LdcDevid, LdcChnId);  
MI\_LDC\_DestroyDevice(LdcDevid);
```

➤ Related APIs

[MI\\_LDC\\_DestroyDevice](#)

## 2.2. MI\_LDC\_DestroyDevice

➤ Description

Destroy an LDC device

➤ Syntax

```
MI_S32 MI_LDC_DestroyDevice(MI_LDC_DEV devId);
```

➤ Parameters

| Parameter Name | Description      | Input/Output |
|----------------|------------------|--------------|
| devId          | Device ID number | Input        |

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

➤ Dependence

- Header file: mi\_ldc.h
- Lib: libmi\_ldc.a

➤ Note

- None.

➤ Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

➤ Related APIs

[MI\\_LDC\\_CreateDevice](#).

## 2.3. MI\_LDC\_CreateChannel

➤ Description

Create an LDC channel

➤ Syntax

```
MI_S32 MI_LDC_CreateChannel(MI_LDC_DEV devId, MI_LDC_CHN chnId,  
MI\_LDC\_ChnAttr\_t *pstChnAttr);
```

➤ Parameters

| Parameter Name | Description        | Input/Output |
|----------------|--------------------|--------------|
| devId          | Device ID number   | Input        |
| chnId          | LDC Channel number | Input        |
| pstChnAttr     | Channel attributes | Input        |

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

➤ Dependence

- Header file: mi\_ldc.h
- Lib: libmi\_ldc.a

➤ Note

This API needs to be called after [MI\\_LDC\\_CreateDevice](#).

➤ Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

➤ Related APIs

[MI\\_LDC\\_DestroyChannel](#).

## 2.4. MI\_LDC\_DestroyChannel

### ➤ Description

Destroy an LDC channel

### ➤ Syntax

```
MI_S32 MI_LDC_DestroyChannel(MI_LDC_DEV devId, MI_LDC_CHN chnId);
```

### ➤ Parameters

| Parameter Name | Description        | Input/Output |
|----------------|--------------------|--------------|
| devId          | Device ID number   | Input        |
| chnId          | LDC Channel number | Input        |

### ➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

### ➤ Dependence

- Header file: mi\_ldc.h
- Lib: libmi\_ldc.a

### ➤ Note

None.

### ➤ Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

### ➤ Related APIs

[MI\\_LDC\\_CreateChannel](#).

## 2.5. MI\_LDC\_StartChannel

### ➤ Description

Start an LDC channel.

### ➤ Syntax

```
MI_S32 MI_LDC_StartChannel(MI_LDC_DEV devId, MI_LDC_CHN chnId);
```

### ➤ Parameters

| Parameter Name | Description        | Input/Output |
|----------------|--------------------|--------------|
| devId          | Device ID number   | Input        |
| chnId          | LDC Channel number | Input        |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see [ERROR CODE](#) for details.
- Dependence
  - Header file: mi\_ldc.h
  - Lib: libmi\_ldc.a
- Note

Call this API after Device and Channel have been created.
- Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.
- Related APIs

[MI\\_LDC\\_StopChannel](#).

## 2.6. MI\_LDC\_StopChannel

- Description

Stop an LDC channel.
- Syntax

MI\_S32 MI\_LDC\_StopChannel(MI\_LDC\_DEV devId, MI\_LDC\_CHN chnId);
- Parameters

| Parameter Name | Description        | Input/Output |
|----------------|--------------------|--------------|
| devId          | Device ID number   | Input        |
| chnId          | LDC Channel number | Input        |

- Return Value
  - Zero: Successful
  - Non-zero: Failed, see [ERROR CODE](#) for details.
- Dependence
  - Header file: mi\_ldc.h
  - Lib: libmi\_ldc.a
- Note

None
- Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

➤ Related APIs

[MI\\_LDC\\_StartChannel](#).

## 2.7. MI\_LDC\_GetOutputPortAttr

➤ Description

Get the LDC output attribute.

➤ Syntax

```
MI_S32 MI_LDC_GetOutputPortAttr(MI_LDC_DEV devId, MI_LDC_CHN chnId,  
MI\_LDC\_OutputPortAttr\_t *pstOutputAttr);
```

➤ Parameters

| Parameter Name | Description        | Input/Output |
|----------------|--------------------|--------------|
| devId          | Device ID number   | Input        |
| chnId          | LDC Channel number | Input        |
| pstOutputAttr  | Output attribute   | Output       |

➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

➤ Dependence

- Header file: `mi_ldc.h`
- Lib: `libmi_ldc.a`

➤ Note

You can obtain the information through this interface after Create Channel.

➤ Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

➤ Related APIs

None

## 2.8. MI\_LDC\_SetChnParam

➤ Description

Set the LDC channel attribute.

## ➤ Syntax

```
MI_S32 MI_LDC_SetChnParam (MI_LDC_DEV devId, MI_LDC_CHN chnId,
MI\_LDC\_ChnParam\_t *pstChnParam);
```

## ➤ Parameters

| Parameter Name | Description                     | Input/Output |
|----------------|---------------------------------|--------------|
| devId          | Device ID number                | Input        |
| chnId          | Channel number                  | Input        |
| pstChnParam    | Channel configuration parameter | Input        |

## ➤ Return Value

- Zero: Successful
- Non-zero: Failed, see [ERROR CODE](#) for details.

## ➤ Dependence

- Header file: `mi_ldc.h`
- Lib: `libmi_ldc.a`

## ➤ Note

- Need to be used after `MI_LDC_CreateChannel`.

## ➤ Example

Used when updating channel configuration, switching algorithm mode and adjusting lens parameters.

```
MI\_LDC\_DEV devId = 0;
MI\_LDC\_CHN chnId = 0;
MI\_LDC\_ChnParam\_t stChnParam = {};
MI\_LDC\_SensorCalibInfo\_t *pCalibInfo = &stChnParam.stCalibInfo;

s32Ret = ReadBufFromFile("ldc.json", &stChnParam.pConfigAddr, &stChnParam.u32ConfigSize);
if (s32Ret)
{
    return s32Ret;
}
s32Ret = ReadBufFromFile("CalibPoly.bin", &pstCalibInfo.pCalibPolyBinAddr,
&pstCalibInfo.u32CalibPolyBinSize);
if (s32Ret)
{
    return s32Ret;
}

MI\_LDC\_SetChnParam(devId, chnId, &stChnParam) ;

ReleaseBuf(stChnParam.pConfigAddr);
ReleaseBuf(pstCalibInfo.pCalibPolyBinAddr);
```

## ➤ Related APIs

None

## 2.9. MI\_LDC\_DoLutDirectTask

### ➤ Description

Execute table lookup tasks , send buffer directly to ldc hardware for processing.

### ➤ Syntax

```
MI_S32 MI_LDC_DoLutDirectTask(MI_LDC_DEV devID, MI_LDC_CHN chnId,
MI_LDC_LutTaskAttr_t *pstAttr);
```

### ➤ Parameters

| Parameter Name | Description                                   | Input/Output |
|----------------|-----------------------------------------------|--------------|
| devID          | Device initialization parameters              | Input        |
| pstAttr        | Table looking-up task configuration attribute | Input/Output |

### ➤ Return Value

- Zero: Successful
- None-zero: Failed, see [ERROR CODE](#) for details.

### ➤ Dependence

- Header : mi\_ldc.h
- Lib : libmi\_ldc.a

### ➤ Note

None

### ➤ Example

```
MI_S32 s32Ret = MI_SUCCESS;
MI_LDC_DEV devId = 0;
MI_LDC_CHN chnId = 0;
MI_LDC_LutTaskAttr_t stLutTask = {};
MI_LDC_DirectBuf_t stTableX = {}, stTableY = {}, stTableWeight = {};
void *pVirTableX = NULL, *pVirTableY=NULL, *pVirTableWeight = NULL;

#define LUT_TABLE_Init(path, table, ppVir, w, h, __exit_func__)
do { \
table.ePixelFormat = E_MI_SYS_PIXEL_FRAME_I8; \
table.u32Width = w; \
table.u32Height = h; \
table.u32Stride[0] = w; \

s32Ret = MI_SYS_MMa_Alloc(0, NULL, w*h, &table.phyAddr[0]); \
if (s32Ret)
{
printf("failed to malloc buf\n");
goto __exit_func__;
}
s32Ret = MI_SYS_Mmap(table.phyAddr[0], w*h, ppVir, false); \
if (s32Ret)
```

```

{
    printf("failed to Mmap buf\n");
    goto __exit_func__;
}
s32Ret = ReadBufFromFile(path, ppVir, w*h);
if (s32Ret)
{
    goto __exit_func__;
}
} while(0)

#define LUT_TABLE_Deinit(table, pVir)
do { \
    if (pVir)
    {
        MI_SYS_Munmap(pVir, table.u32Width * table.u32Height);
    }
    if (table.phyAddr[0])
    {
        MI_SYS_MMA_Free(0, table.phyAddr[0]);
    }
} while(0)

LUT_TABLE_Init("1080p_tableX", stTableX, & pVirTableX, 1920, 1080, __exit);
LUT_TABLE_Init("1080p_tableY", stTableY, & pVirTableY, 1920, 1080, __exit);
LUT_TABLE_Init("1080p_tableWeight", stTableWeight, &pVirTableWeight, 1920, 1080, __exit);

MI\_LDC\_DoLutDirectTask (devId, chnId, & stLutTask) ;

__exit:
LUT_TABLE_Deinit(stTableX, pVirTableX);
LUT_TABLE_Deinit(stTableY, pVirTableY);
LUT_TABLE_Deinit(stTableWeight, pVirTableWeight);

```

- Related Type  
None

## 2.10. [MI\\_LDC\\_SetOutputPortAttr](#)

- Description  
Configure LDC output attributes.

- Syntax  
  
MI\_S32 MI\_LDC\_SetOutputPortAttr(MI\_LDC\_DEV devId, MI\_LDC\_CHN chnId,  
[MI\\_LDC\\_OutputPortAttr\\_t](#) \*pstOutputAttr);

- Parameters

| Parameter Name | Description | Input/Output |
|----------------|-------------|--------------|
| devId          | Device id   | Input        |



| Parameter Name | Description      | Input/Output |
|----------------|------------------|--------------|
| chnId          | Ldc channel id   | Input        |
| pstOutputAttr  | Output attribute | Input        |

➤ Return Value

- Zero: Successful
- None-zero: Failed, see [ERROR CODE](#) for details.

➤ Dependence

- Header : mi\_ldc.h
- Lib : libmi\_ldc.a

➤ Note

1. Used it after the channel is created (MI\_LDC\_CreateChannel).
2. Only support to configure the format outputted by LDC, supported formats: NV12, ARGB8888, BGRA8888.
3. Tiramisu don't Support this interface

➤ Example

```
MI_LDC_DEV devId = 0;
MI_LDC_CHN chnId = 0;
MI_LDC_OutputPortAttr_t stLdcPortAttr = {};

MI_LDC_GetOutputPortAttr(devId, chnId, &stLdcPortAttr);
stLdcPortAttr.ePixelFormat = E_MI_SYS_PIXEL_FRAME_ARGB8888;
MI_LDC_SetOutputPortAttr(devId, chnId, &stLdcPortAttr);
```

➤ Related Type

None

## 2.11. MI\_LDC\_AttachToChn

➤ Description

Attach Ldc Channel To Another Module Channel.

➤ Syntax

```
MI_S32 MI_LDC_AttachToChn(MI_LDC_DEV devId, MI_LDC_CHN chnId,
MI_SYS_ChnPort_t *pstChnPort);
```

➤ Parameters

| Parameter Name | Description              | Input/Output |
|----------------|--------------------------|--------------|
| devId          | Device id                | Input        |
| chnId          | Ldc channel id           | Input        |
| pstChnPort     | Channel port to be bound | Input        |

- Return Value
  - Zero: Successful
  - None-zero: Failed, see [ERROR CODE](#) for details.
- Dependence
  - Header : mi\_ldc.h
  - Lib : libmi\_ldc.a
- Note
  1. Used it after the channel is created (MI\_LDC\_CreateChannel).
  2. Only binding ISP modules is supported.
- Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.
- Related Type

None

## 2.12. MI\_LDC\_DetachFromChn

- Description

Detach Ldc Channel.
- Syntax

```
MI_S32 MI_LDC_DetachFromChn(MI_LDC_DEV devId, MI_LDC_CHN chnId,
MI_SYS_ChnPort_t *pstChnPort);
```

- Parameters

| Parameter Name | Description              | Input/Output |
|----------------|--------------------------|--------------|
| devId          | Device id                | Input        |
| chnId          | Ldc channel id           | Input        |
| pstChnPort     | Channel port to be bound | Input        |

- Return Value
  - Zero: Successful
  - None-zero: Failed, see [ERROR CODE](#) for details.
- Dependence
  - Header : mi\_ldc.h
  - Lib : libmi\_ldc.a
- Note

None

➤ Example

Refer to [MI\\_LDC\\_CreateDevice](#) example.

➤ Related Type

None

### 3. LDC DATA TYPE

---

LDC module related data types are listed below:

| Data type                                    | Definition                                 |
|----------------------------------------------|--------------------------------------------|
| <a href="#">MI_LDC_DEV</a>                   | Define LDC device type.                    |
| <a href="#">MI_LDC_CHN</a>                   | Define LDC channel type.                   |
| <a href="#">MI_LDC_DevAttr_t</a>             | Define LDC device attribute.               |
| <a href="#">MI_LDC_WorkMode_e</a>            | Define LDC work mode.                      |
| <a href="#">MI_LDC_MapInfoType_e</a>         | Define LDC mapping information type.       |
| <a href="#">MI_LDC_DisplacementMapInfo_t</a> | Define LDC displacement map information.   |
| <a href="#">MI_LDC_SensorCalibInfo_t</a>     | Define LDC Sensor calibration information. |
| <a href="#">MI_LDC_ChnAttr_t</a>             | Define LDC channel attribute.              |
| <a href="#">MI_LDC_ChnParam_t</a>            | Define LDC channel dynamic parameters.     |
| <a href="#">MI_LDC_OutputPortAttr_t</a>      | Define LDC output port attribute.          |
| <a href="#">MI_LDC_DirectBuf_t</a>           | Define LDC input buffer type.              |
| <a href="#">MI_LDC_LutTaskAttr_t</a>         | Define LDC image lookup table task type.   |

#### 3.1. MI\_LDC\_DEV

➤ Description

Define LDC device type

➤ Definition

```
typedef MI_U32 MI_LDC_DEV;
```

➤ Note

None

➤ Related Type

None

## 3.2. MI\_LDC\_CHN

- Description  
Define LDC channel type
- Definition  

```
typedef MI_U32 MI_LDC_CHN;
```
- Note  
None
- Related Type  
None

## 3.3. MI\_LDC\_DevAttr\_t

- Description  
LDC device initialization Parameters.
- Definition  

```
typedef struct MI_LDC_DevAttr_s  
{  
    MI_U32 u32Reserved;  
} MI_LDC_DevAttr_t;
```
- Member

| Member name | Description    |
|-------------|----------------|
| u32Reserved | Reserved Value |
- Note  
None
- Related Type  
[MI\\_LDC\\_CreateDevice](#)

## 3.4. MI\_LDC\_WorkMode\_e

- Description  
Define LDC work mode.

## ➤ Definition

```
typedef enum
{
    MI_LDC_WORKMODE_LDC = 0x01,
    MI_LDC_WORKMODE_LUT = 0x02,
    MI_LDC_WORKMODE_DIS_GYRO = 0x04,
} MI_LDC_WorkMode_e;
```

## ➤ Member

| Member name              | Description                            |
|--------------------------|----------------------------------------|
| MI_LDC_WORKMODE_LDC      | LDC mode                               |
| MI_LDC_WORKMODE_LUT      | LUT direct lookup mode                 |
| MI_LDC_WORKMODE_DIS_GYRO | DIS anti-shake mode based on gyroscope |

## ※ Note

None

## ➤ Related Type

[MI\\_LDC\\_ChnAttr\\_t](#)

### 3.5. MI\_LDC\_MapInfoType\_e

## ➤ Description

LDC mapping information type.

## ➤ Definition

```
typedef enum
{
    MI_LDC_MAPINFOTYPE_DISPMAP,
    MI_LDC_MAPINFOTYPE_SENSORCALIB,
} MI_LDC_MapInfoType_e ;
```

## ➤ Member

| Member name                    | Description                    |
|--------------------------------|--------------------------------|
| MI_LDC_MAPINFOTYPE_DISPMAP     | Displacement map               |
| MI_LDC_MAPINFOTYPE_SENSORCALIB | sensor calibration information |

## ※ Note

None

➤ Related Type

[MI\\_LDC\\_ChnAttr\\_t](#)

### 3.6. MI\_LDC\_DisplacementMapInfo\_t

➤ Description

LDC displacement map information.

➤ Definition

```
typedef struct MI_LDC_DisplacementMapInfo_s
{
    void *pXmapAddr;
    void *pYmapAddr;
    MI_U32 u32XmapSize;
    MI_U32 u32YmapSize;
} MI_LDC_DisplacementMapInfo_t;
```

➤ Member

| Member name | Description                                            |
|-------------|--------------------------------------------------------|
| pXmapAddr   | The starting address of the X coordinate mapping table |
| pYmapAddr   | The starting address of the Y coordinate mapping table |
| u32XmapSize | X coordinate mapping table size                        |
| u32YmapSize | Y coordinate mapping table size                        |

※ Note

None

➤ Related Type

[MI\\_LDC\\_ChnAttr\\_t](#)

[MI\\_LDC\\_ChnParam\\_t](#)

### 3.7. MI\_LDC\_SensorCalibInfo\_t

➤ Description

LDC Sensor calibration information.

➤ Definition

```
typedef struct MI_LDC_SensorCalibInfo_s
{
    void *pCalibPolyBinAddr;
    MI_U32 u32CalibPolyBinSize;
} MI_LDC_SensorCalibInfo_t;
```

➤ Member

| Member name         | Description                     |
|---------------------|---------------------------------|
| pCalibPolyBinAddr   | Calibration information pointer |
| u32CalibPolyBinSize | Calibration message size        |

※ Note

None

➤ Related Type

[MI\\_LDC\\_ChnAttr\\_t](#)

[MI\\_LDC\\_ChnParam\\_t](#)

### 3.8. MI\_LDC\_ChnAttr\_t

➤ Description

LDC channel attribute.

➤ Definition

```
typedef struct MI_LDC_ChnAttr_s
{
    MI\_LDC\_WorkMode\_e eMode;

    MI_BOOL bUseProjection3x3Matrix;
    MI_S32 as32Projection3x3Matrix[LDC_MAXTRIX_NUM];
    MI_U16 u16UserSliceNum;
    MI_U16 u16FocalLength;

    void *pConfigAddr;
    MI_U32 u32ConfigSize;

    MI\_LDC\_MapInfoType\_e eInfoType;
    union
```



```

{
    MI\_LDC\_DisplacementMapInfo\_t stDispMapInfo;
    MI\_LDC\_SensorCalibInfo\_t stCalibInfo;
};
} MI_LDC_ChnAttr_t;

```

➤ Member

| Member name               | Description                                                                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eMode                     | Channel working mode                                                                                                                                                                                                                       |
| bUseProjection3x3Matrix   | Indicates whether to use as32Projection3x3Matrix                                                                                                                                                                                           |
| as32Projection3x3Matrix   | 3x3 Matrix information indicates the direction and position of the gyroscope (used in DIS mode)                                                                                                                                            |
| u16UserSliceNum           | The number of slices in the vertical direction, mainly used in the motion scene of the Rolling Shutter camera, each slice will have a separate 3x3 Matrix                                                                                  |
| u16CropRatio              | The crop ratio of DIS output image.<br>For example, the input image is 1920x1080 and configure the crop ratio to 80, then the output image will be cropped $(1-80\%)/2 = 10\%$ , the output image is 1536x864, which is the aligned image. |
| u16FocalLength            | Focal length, unit: mm (used in DIS mode)                                                                                                                                                                                                  |
| pConfigAddr               | LDC configuration information address                                                                                                                                                                                                      |
| u32ConfigSize             | LDC configuration information size                                                                                                                                                                                                         |
| eInfoType                 | LDC mapping information type                                                                                                                                                                                                               |
| stDispMapInfo/stCalibInfo | LDC displacement table/sensor calibration information, includes distortion parameters                                                                                                                                                      |

➤ Note

None

➤ Related Type

[MI\\_LDC\\_CreateChannel](#)

### 3.9. [MI\\_LDC\\_ChnParam\\_t](#)

➤ Description

Define LDC channel dynamic parameters.

➤ Definition

```
typedef struct MI_LDC_ChnParam_s
{
    MI_BOOL bUseProjection3x3Matrix;
    MI_S32 as32Projection3x3Matrix[LDC_MAXTRIX_NUM];
    MI_U16 u16FocalLength;
    void *pConfigAddr;
    MI_U32 u32ConfigSize;
    union
    {
        MI\_LDC\_DisplacementMapInfo\_t stDispMapInfo;
        MI\_LDC\_SensorCalibInfo\_t stCalibInfo;
    };
} MI_LDC_ChnParam_t;
```

➤ Member

| Member name                | Description                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------|
| bUseProjection3x3Matrix    | Indicates whether to use as32Projection3x3Matrix                                                |
| as32Projection3x3Matrix    | 3x3 Matrix information indicates the direction and position of the gyroscope (used in DIS mode) |
| u16FocalLength             | Focal length, unit: mm (used in DIS mode)                                                       |
| pConfigAddr                | LDC configuration information address                                                           |
| u32ConfigSize              | LDC configuration information size                                                              |
| stDispMapInfo/ stCalibInfo | LDC displacement table/sensor calibration information, includes distortion parameters           |

※ Note

None

➤ Related Type

[MI\\_LDC\\_SetChnParam](#)

### 3.10. [MI\\_LDC\\_OutputPortAttr\\_t](#)

➤ Description

Define LDC output port attribute

➤ Definition

```
typedef struct MI_LDC_OutputPortAttr_s
{
    MI_U16 u16Width;
    MI_U16 u16Height;
    MI_SYS_PixelFormat_e ePixelFormat;
} MI_LDC_OutputPortAttr_t;
```

➤ Member

| Member name  | Description        |
|--------------|--------------------|
| u16Width     | Image width        |
| u16Height    | Image height       |
| ePixelFormat | Image pixel format |

➤ Note

None

➤ Related Type

[MI\\_LDC\\_GetOutputPortAttr](#)

### 3.11. MI\_LDC\_DirectBuf\_t

➤ Description

LDC input buffer type.

➤ Definition

```
typedef struct MI_LDC_DirectBuf_s
{
    MI_SYS_PixelFormat_e ePixelFormat;
    MI_U32 u32Width;
    MI_U32 u32Height;
    MI_U32 u32Stride[2];
    MI_PHY phyAddr[2];
} MI_LDC_DirectBuf_t;
```

➤ Member

| Member name  | Description                    |
|--------------|--------------------------------|
| ePixelFormat | Image format                   |
| u32Width     | Image width                    |
| u32Height    | Image height                   |
| u32Stride    | Image occupied bytes per line  |
| phyAddr      | Image storage physical address |

※ Note

None

➤ Related Type

[MI\\_LDC\\_LutTaskAttr\\_t](#)

### 3.12. MI\_LDC\_LutTaskAttr\_t

➤ Description

LDC image lookup table task type.

➤ Definition

```
typedef struct MI_LDC_LutTaskAttr_s
{
    struct MI_LDC_LutSrcBuf_s
    {
        MI_LDC_DirectBuf_t stTableX;
        MI_LDC_DirectBuf_t stTableY;
        MI_LDC_DirectBuf_t stTableWeight;
    } stSrcBuf;

    MI_LDC_DirectBuf_t stDstBuf;
} MI_LDC_LutTaskAttr_t;
```

➤ Member

| Member name | Description                         |
|-------------|-------------------------------------|
| stSrcBuf    | Table data source and weight lookup |
| stDstBuf    | Store the out data                  |

※ Note

None

➤ Related Type

[MI\\_LDC\\_DoLutDirectTask](#)

## 4. ERROR CODE

---

LDC API error code is defined in the table below:

Table 4-1: LDC API Error Code

| Error Code | Macro Definition            | Description                    |
|------------|-----------------------------|--------------------------------|
| 0xA0172001 | MI_ERR_LDC_INVALID_DEVID    | Invalid device id              |
| 0xA0172002 | MI_ERR_LDC_INVALID_CHNID    | Invalid channel id             |
| 0xA0172003 | MI_ERR_LDC_ILLEGAL_PARAM    | Illegal Input argument         |
| 0xA0172006 | MI_ERR_LDC_NULL_PTR         | Illegal null pointer argument  |
| 0xA0172008 | MI_ERR_LDC_NOT_SUPPORT      | No support feature             |
| 0xA017200C | MI_ERR_LDC_NOT_NOMEM        | Not enough memory              |
| 0xA0172012 | MI_ERR_LDC_BUSY             | Channel is busy                |
| 0xA0172015 | MI_ERR_LDC_MOD_NOT_INIT     | Module is not inited           |
| 0xA0172016 | MI_ERR_LDC_MOD_INITED       | Module has been inited         |
| 0xA0172018 | MI_ERR_LDC_PORT_NOT_DISABLE | Port is not be disable         |
| 0xA017201F | MI_ERR_LDC_FAIL             | Ldc failed to exec             |
| 0xA0172300 | MI_ERR_LDC_DEV_CREATED      | Device has been created        |
| 0xA0172301 | MI_ERR_LDC_DEV_NOT_CREATE   | Device has not been created    |
| 0xA0172302 | MI_ERR_LDC_DEV_NOT_DESTROY  | Device has not been destroyed  |
| 0xA0172303 | MI_ERR_LDC_CHN_CREATED      | Channel has been created       |
| 0xA0172304 | MI_ERR_LDC_CHN_NOT_CREATE   | Channel has not been created   |
| 0xA0172305 | MI_ERR_LDC_CHN_NOT_STOP     | Channel has not been stopped   |
| 0xA0172306 | MI_ERR_LDC_CHN_NOT_DESTROY  | Channel has not been destroyed |
| 0xA0172307 | MI_ERR_LDC_PORT_NOT_UNBIND  | Port has not been unbinded     |

## 5. PROCFS INTRODUCTION

### 5.1. cat

- Debug info  
# cat /proc/mi\_modules/mi\_ldc/mi\_ldc0
- Debug info analysis  
Record LDC current usage status and relative attributes, which can be dynamically got for debug and test.
- Parameter Description

```
----- start dump LDC DEV info -----
DevID  IRQ_Enable  IsrCnt  IsrNum  TotBlkNum  FramDonCnt  cmdqif
  0      1      553    52      330         1         0
HalInFrmCnt HalDoneFrmCnt HalDoneBlkCnt HalIsrDoneCnt HalIsrErrCnt
      2          2          60          555          0
----- End dump LDC dev 0 info -----
```

| Parameter   |               | Description                                             |
|-------------|---------------|---------------------------------------------------------|
| Device Info | DevID         | Device ID                                               |
|             | IRQ_Enable    | IRQ enable state                                        |
|             | IsrCnt        | Count of interrupts, cmdq mode has no reference meaning |
|             | IsrNum        | Interrupt number                                        |
|             | TotBlkNum     | Total number of ldc blocks that need to be processed    |
|             | FramDonCnt    | Number of processed frames                              |
|             | cmdqif        | cmdq interface address                                  |
|             | HalInFrmCnt   | Total number of frames input to Hal                     |
|             | HalDoneFrmCnt | Total number of frames processed by Hal                 |
|             | HalDoneBlkCnt | Total number of blocks processed by Hal                 |
|             | HalIsrDoneCnt | Number of Hal done interrupts                           |
|             | HalIsrErrCnt  | Number of Hal Err interrupts                            |

```

----- start dump LDC CHN info -----
ChnId  workmode  status  InputWH  pixel  stride  InBufCnt  InTodoCnt  atom
  0      ldc      1(1920x1080)  11    1920      2          2        0
EptzInit  OneBinSz    OneBin  TotalBin  LdcDone  EnqTDeq  BlendT
19512us   331071us    595343us 1901949us 466484us 499099us  0us

```

| Parameter    |           | Description                                                     |
|--------------|-----------|-----------------------------------------------------------------|
| Channel Info | ChnId     | Channel ID                                                      |
|              | workmode  | Work mode of current channel                                    |
|              | status    | Status of current channel:<br>0:init 1:start 2:stop 3:destroyed |
|              | InputWH   | Input frame resolution                                          |
|              | Pixel     | Input frame format                                              |
|              | Stride    | Input frame stride[0]                                           |
|              | InBufCnt  | The total number of frames input to the pre-level               |
|              | IntodoCnt | The label of the frame to be processed (incremental)            |
|              | Atom      | The number of buffers held by the bottom layer                  |
|              | EptzInit  | Time used to initialize the algorithm                           |
|              | OneBinSz  | Time used to calculate the buffer size of the latest view       |
|              | OneBin    | Time used to calculate the latest view                          |
|              | TotalBin  | Time used to calculate all views by LDC algorithm               |
|              | LdcDone   | LDC hardware processing time                                    |
|              | EnqTDeq   | Total time from getting the pre-level buf to processing it      |
|              | BlendT    | Alpha blending time                                             |

```
LDC Table Info:
ChnId  viewId  mode  NumRuns  ImgFmt  OpMode  flip
0      0      2P_CM  165      0      3      0
      InSize  ViewSize  OutSize  ViewTime
(0000,0000,1920,1080)(0000,0000,1920,0544)(0000,0000,1920,1088)( 219954us)

      InBlk      OutBlk      SclBlk      XYScale      DeltaXY      Crop-En  BlkTime
(1568,0536,0096,0114)(0000,0000,0064,0064)(0000,0000,0064,0064)(0001,0001)(0004,0004)(0001)(0001356us)
(1536,0638,0128,0114)(0000,0000,0064,0064)(0064,0000,0064,0064)(0001,0001)(0004,0004)(0001)(0001286us)
(1472,0734,0128,0112)(0000,0000,0064,0064)(0128,0000,0064,0064)(0001,0001)(0004,0004)(0001)(0001230us)
(1376,0822,0160,0104)(0000,0000,0064,0064)(0192,0000,0064,0064)(0001,0001)(0004,0004)(0001)(0001197us)
(1216,0898,0224,0114)(0000,0000,0096,0064)(0256,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001236us)
(1024,0980,0224,0074)(0000,0000,0096,0064)(0352,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001680us)
(0800,1012,0256,0044)(0000,0000,0096,0064)(0448,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001095us)
(0576,0950,0304,0000)(0000,0000,0096,0064)(0544,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0352,0888,0352,0000)(0000,0000,0096,0064)(0640,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0128,0826,0352,0000)(0000,0000,0096,0064)(0768,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0764,0352,0000)(0000,0000,0096,0064)(0896,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0600,0352,0000)(0000,0000,0096,0064)(1024,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0436,0352,0000)(0000,0000,0096,0064)(1152,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0272,0352,0000)(0000,0000,0096,0064)(1280,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0108,0352,0000)(0000,0000,0096,0064)(1408,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
(0000,0000,0352,0000)(0000,0000,0096,0064)(1536,0000,0096,0064)(0001,0001)(0004,0004)(0001)(0001033us)
```

| Parameter |          | Description                                                             |
|-----------|----------|-------------------------------------------------------------------------|
|           | ChnId    | Channel ID                                                              |
|           | ViewId   | Ldc corresponding Id                                                    |
|           | mode     | Scene mode used by ldc                                                  |
|           | NumRuns  | Total number of blocks that ldc current scene need to run               |
|           | ImgFmt   | Image format processed by ldc                                           |
|           | OpMode   | Current bbox mode:<br>0:2048x16 1:1024x32 2:512x64 3:256x128 4: 128x256 |
|           | flip     | Ldc mirror mode                                                         |
|           | InSize   | The input area of the current view                                      |
|           | ViewSize | The output area of the current view                                     |
|           | OutSize  | The output area of the current scene                                    |
|           | ViewTime | Processing time of the current view                                     |
|           | InBlk    | The input area of the current blk                                       |
|           | OutBlk   | Algorithm output area of current blk                                    |
|           | SclBlk   | The output area of the current blk                                      |
|           | DeltaXY  | The coordinates of the current blk corresponding to the delta table     |
|           | BlkTime  | Corresponding to the processing time of blk                             |

## 5.2. echo

Echo help view available commands:

```
# echo help > /proc/mi_modules/mi_ldc/mi_ldc0
```

|           |                                                                      |
|-----------|----------------------------------------------------------------------|
| Function  | Dump LDC output file                                                 |
| Command   | echo writeoutfile [chnid cnt path] > /proc/mi_modules/mi_ldc/mi_ldc0 |
| Parameter | chnid: channel Id                                                    |



|             |                                                               |
|-------------|---------------------------------------------------------------|
| Description | cnt: dump count                                               |
|             | path: dump path                                               |
| Example     | echo writeoutfile 0 2 /mnt/ > /proc/mi_modules/mi_ldc/mi_ldc0 |

|                       |                                                                                 |
|-----------------------|---------------------------------------------------------------------------------|
| Function              | Dump LDC map file                                                               |
| Command               | echo dumptablebin [chnid view/all path] > /proc/mi_modules/mi_ldc/mi_ldc0       |
| Parameter Description | chnid: channel id                                                               |
|                       | view/all: download mode, view: concrete viewed, all: all views of current scene |
|                       | path: dump path                                                                 |
| Example               | echo dumptablebin 0 all /mnt/ > /proc/mi_modules/mi_ldc/mi_ldc0                 |

|                       |                                                                         |
|-----------------------|-------------------------------------------------------------------------|
| Function              | Configure output properties                                             |
| Command               | echo setout[chnid width height pixel] > /proc/mi_modules/mi_ldc/mi_ldc0 |
| Parameter Description | chnid: channel id                                                       |
|                       | width: output image width                                               |
|                       | height: output image height                                             |
| Example               | echo setout 0 1920 1080 11 > /proc/mi_modules/mi_ldc/mi_ldc0            |

|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| Function              | Loop the previous frame                                    |
| Command               | echo loopbuf [chnid cnt] > /proc/mi_modules/mi_ldc/mi_ldc0 |
| Parameter Description | chnid: channel id                                          |
|                       | cnt: The number of times to loop the previous frame        |
| Example               | echo loopbuf 0 11 > /proc/mi_modules/mi_ldc/mi_ldc0        |

|                       |                                                                                  |
|-----------------------|----------------------------------------------------------------------------------|
| Function              | Configure ldc internal debug trace                                               |
| Command               | echo debug [chnid str] > /proc/mi_modules/mi_ldc/mi_ldc0                         |
| Parameter Description | chnid: channel id                                                                |
|                       | str: debug mode, off: close, ViewRes, ViewLock,ChnRdLock,ChnWrLock,DisGyro, Eptz |
| Example               | echo debug 0 ChnRdLock > /proc/mi_modules/mi_ldc/mi_ldc0                         |

|                       |                                                          |
|-----------------------|----------------------------------------------------------|
| Function              | Switch between software eis and hardware eis             |
| Command               | echo eisswitch [chnid] > /proc/mi_modules/mi_ldc/mi_ldc0 |
| Parameter Description | chnid: channel id                                        |

|         |                                                    |
|---------|----------------------------------------------------|
| Example | echo eisswitch 0 > /proc/mi_modules/mi_ldc/mi_ldc0 |
|---------|----------------------------------------------------|

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| Function                 | Configure a single ldc map                                 |
| Command                  | echo setbin [chnid path] > /proc/mi_modules/mi_ldc/mi_ldc0 |
| Parameter<br>Description | chnid: channel id                                          |
|                          | path: map path                                             |
| Example                  | echo setbin /mnt/ldc.bin > /proc/mi_modules/mi_ldc/mi_ldc0 |